Evolving Memory Architectures for AI

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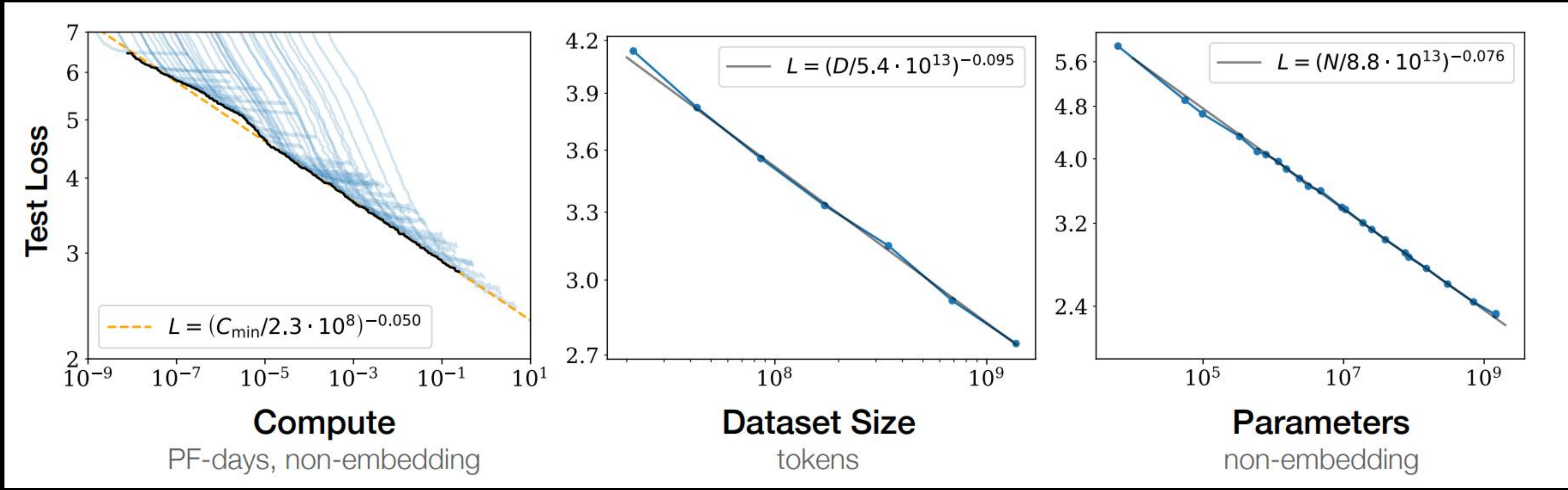
Compute Efficient Frontier

Memory central to continuous scaling and improvement in LLM's



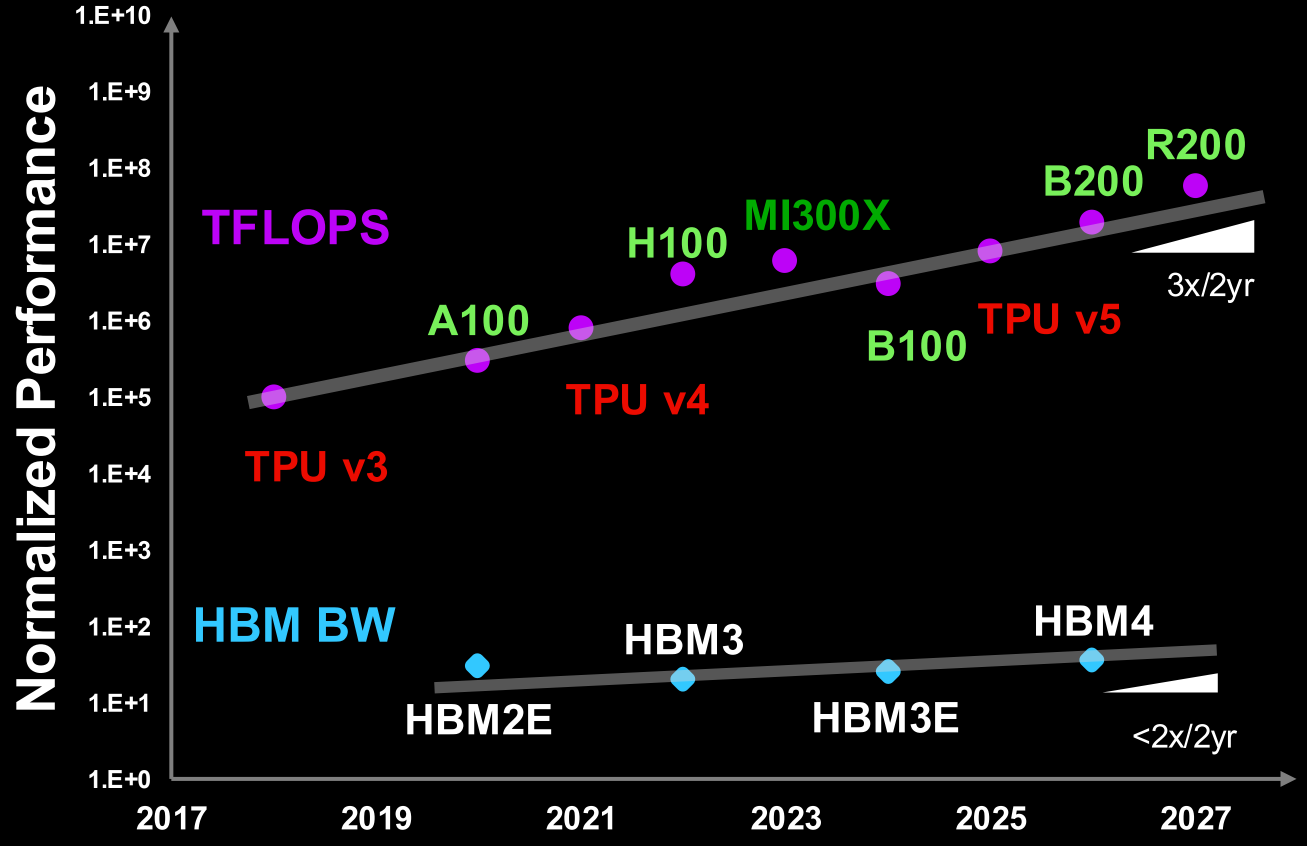
"Language modeling ***performance improves smoothly as we increase the model size, dataset size, and amount of compute used for training****. For optimal performance, all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two."*

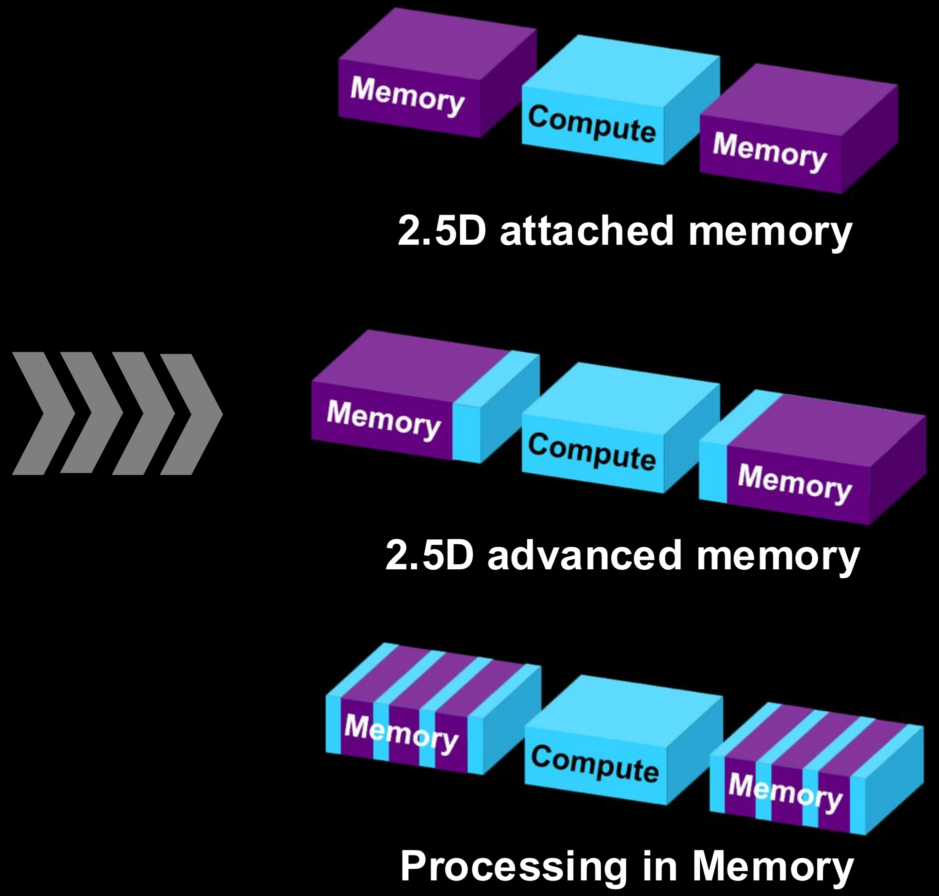
Kaplan et al, OpenAI

Scaling Laws for Neural Language Models, Kaplan, et al, OpenAI

Memory wall

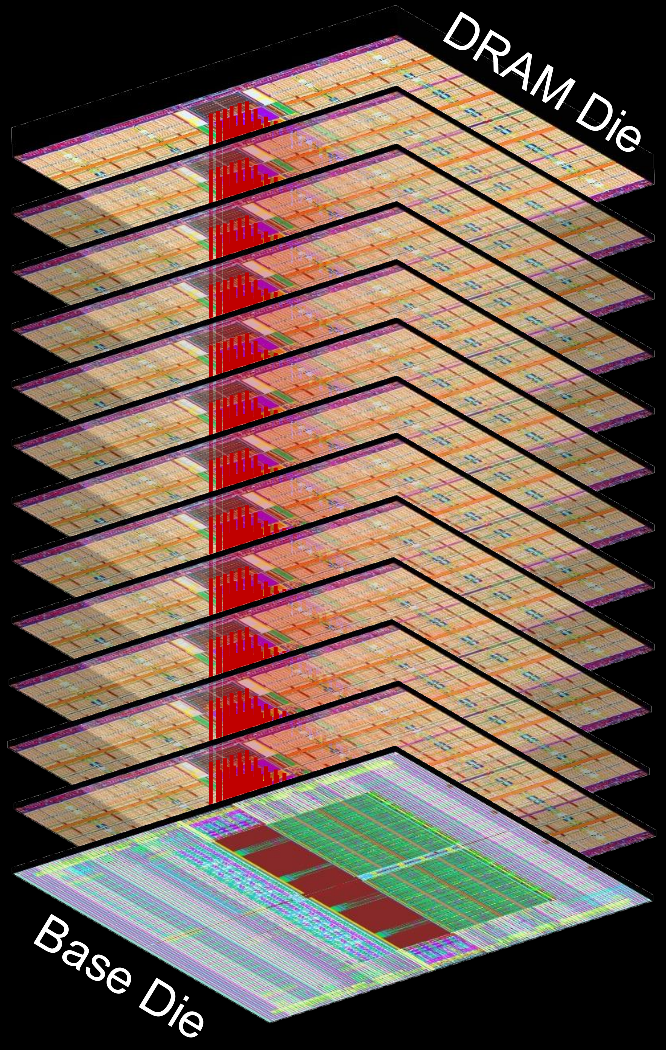
Memory technology has become a key factor limiting AI system performance

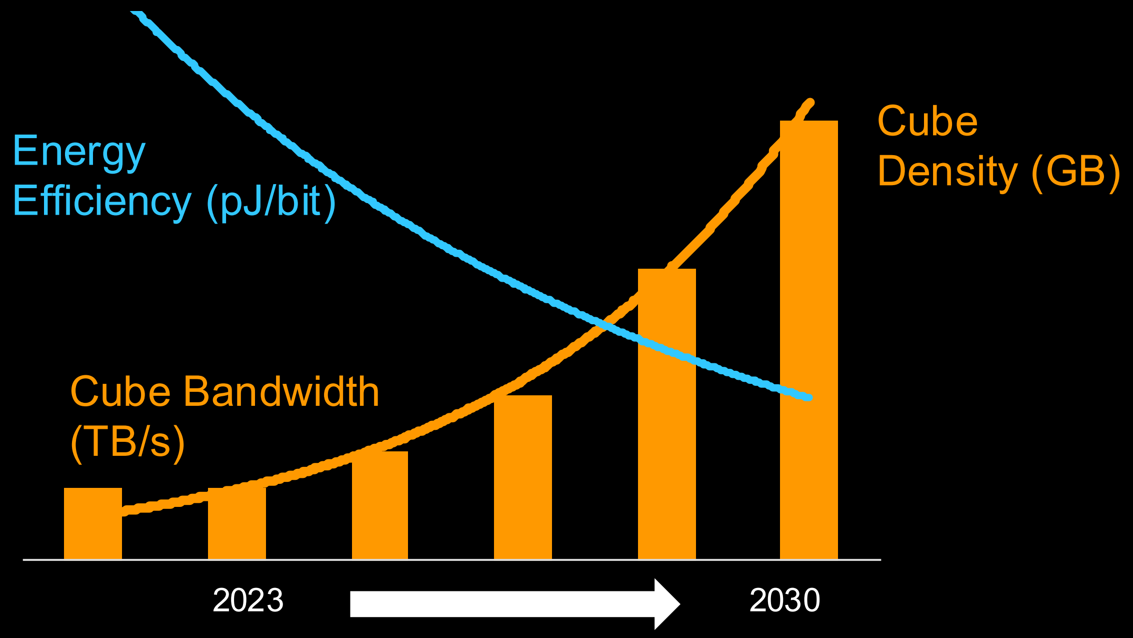




HBM as a compute-memory bridge

Innovations turbocharging AI workloads

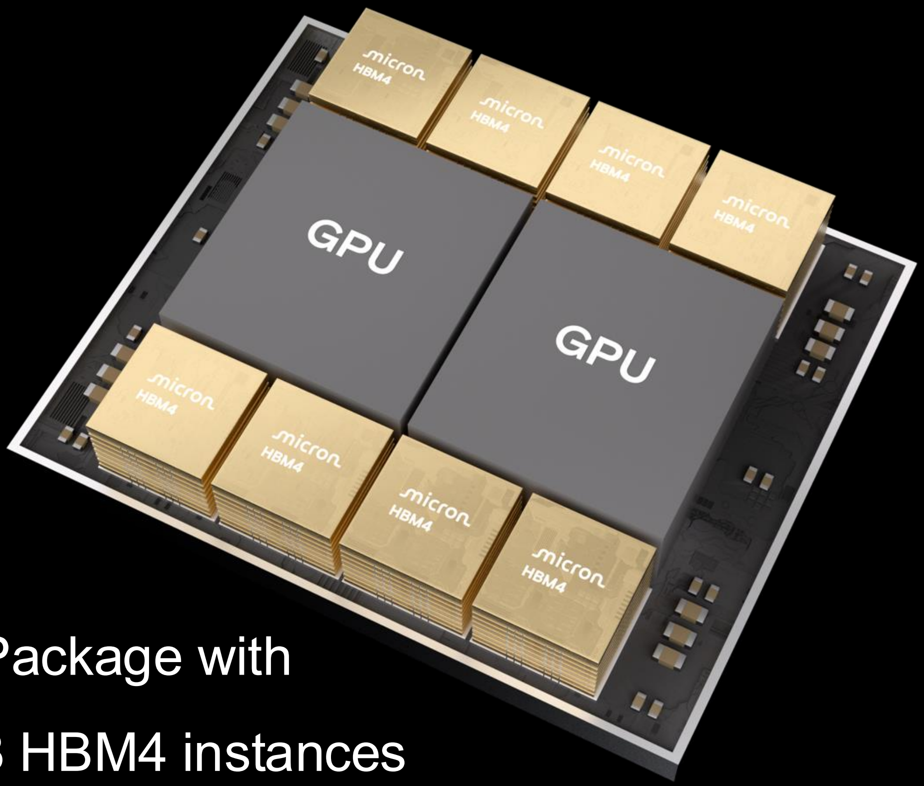




- Major inflections in memory products ahead
- Disruptive process and design innovation
- Packaging technology - next generation capability
- GPU offloading; Advanced D2D PHY
- Custom features

High bandwidth memory - Key enabler of AI performance

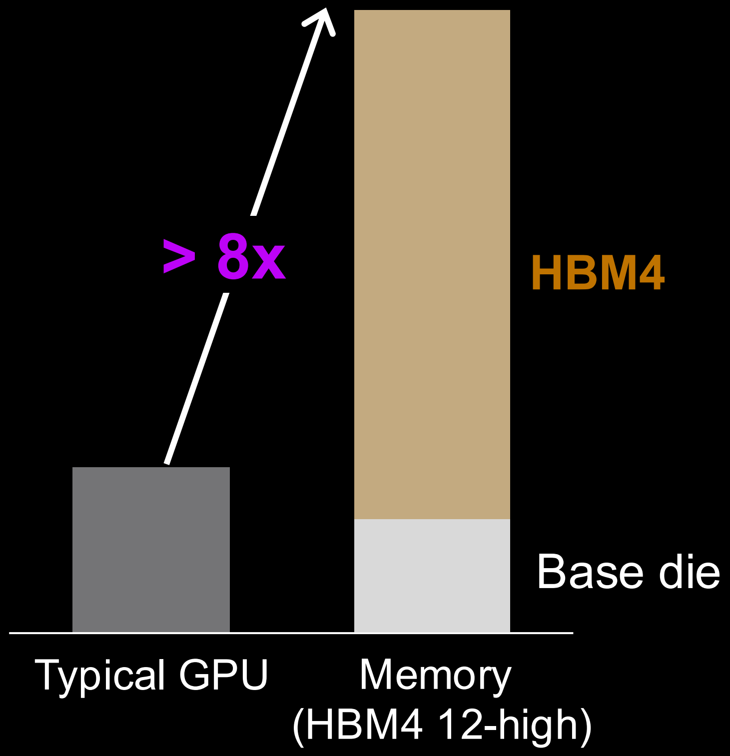
Significant amount of advanced memory Silicon incorporated in leading edge AI platforms



Package with

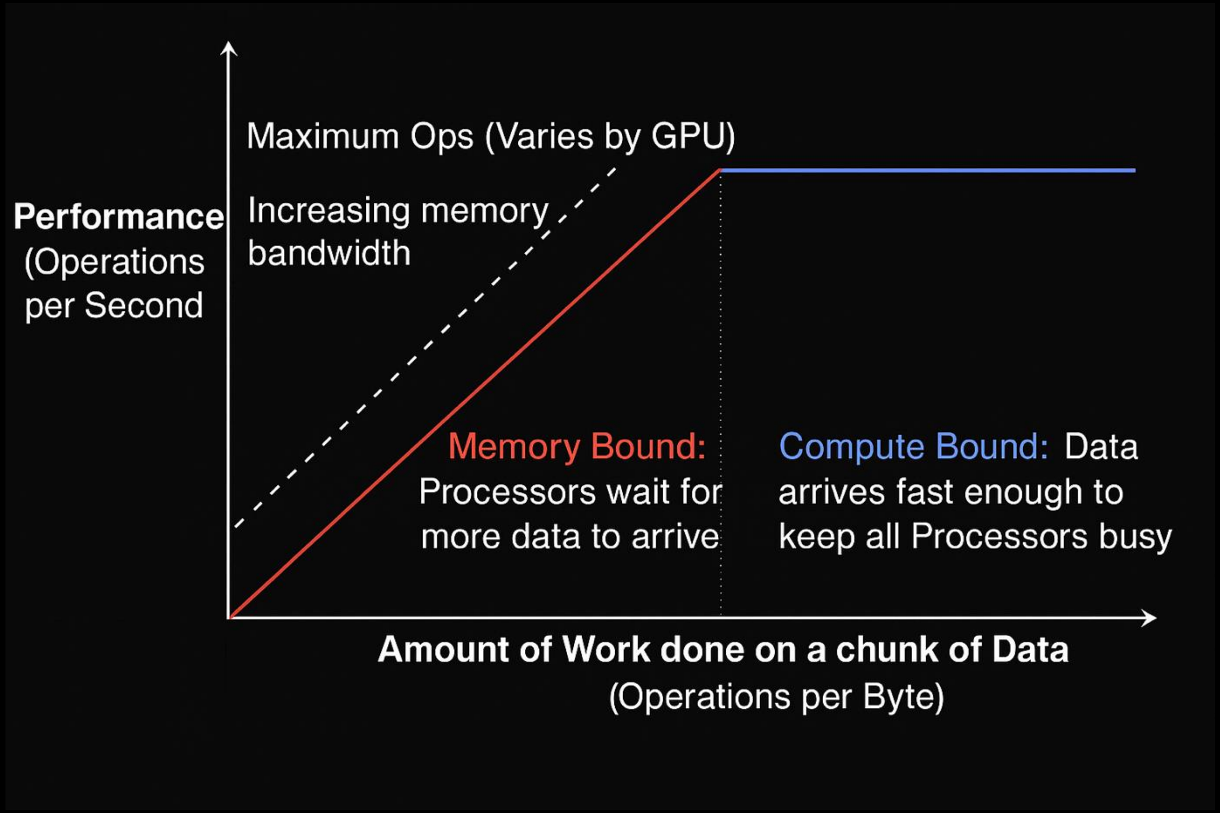
8 HBM4 instances

GPU vs. memory silicon area (mm^2)



Why High Bandwidth Memory (HBM)?

Roofline plot identifies compute vs. memory bottlenecks

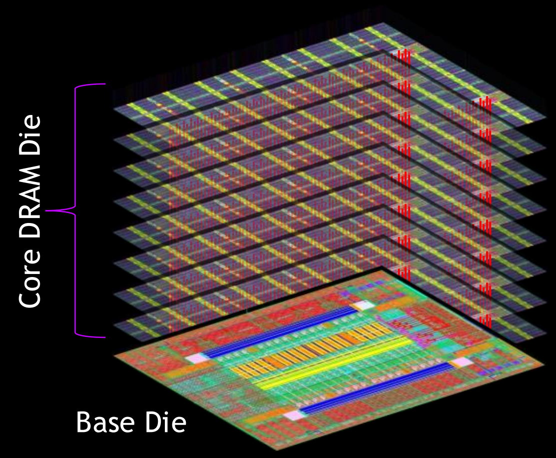


- The model plots attainable performance (OPs) against arithmetic intensity (OPs/byte)
- Applications in the red zone are memory-bound (where memory bandwidth limits performance), while those in the blue zone are compute-bound (where compute capability limits performance)
- HBM elevates the memory bandwidth ceiling, shifting the red zone up, allowing memory-intensive AI workloads to achieve higher performance before hitting memory limits

What is HBM?

HBM is the memory powering accelerated computing behind the AI revolution

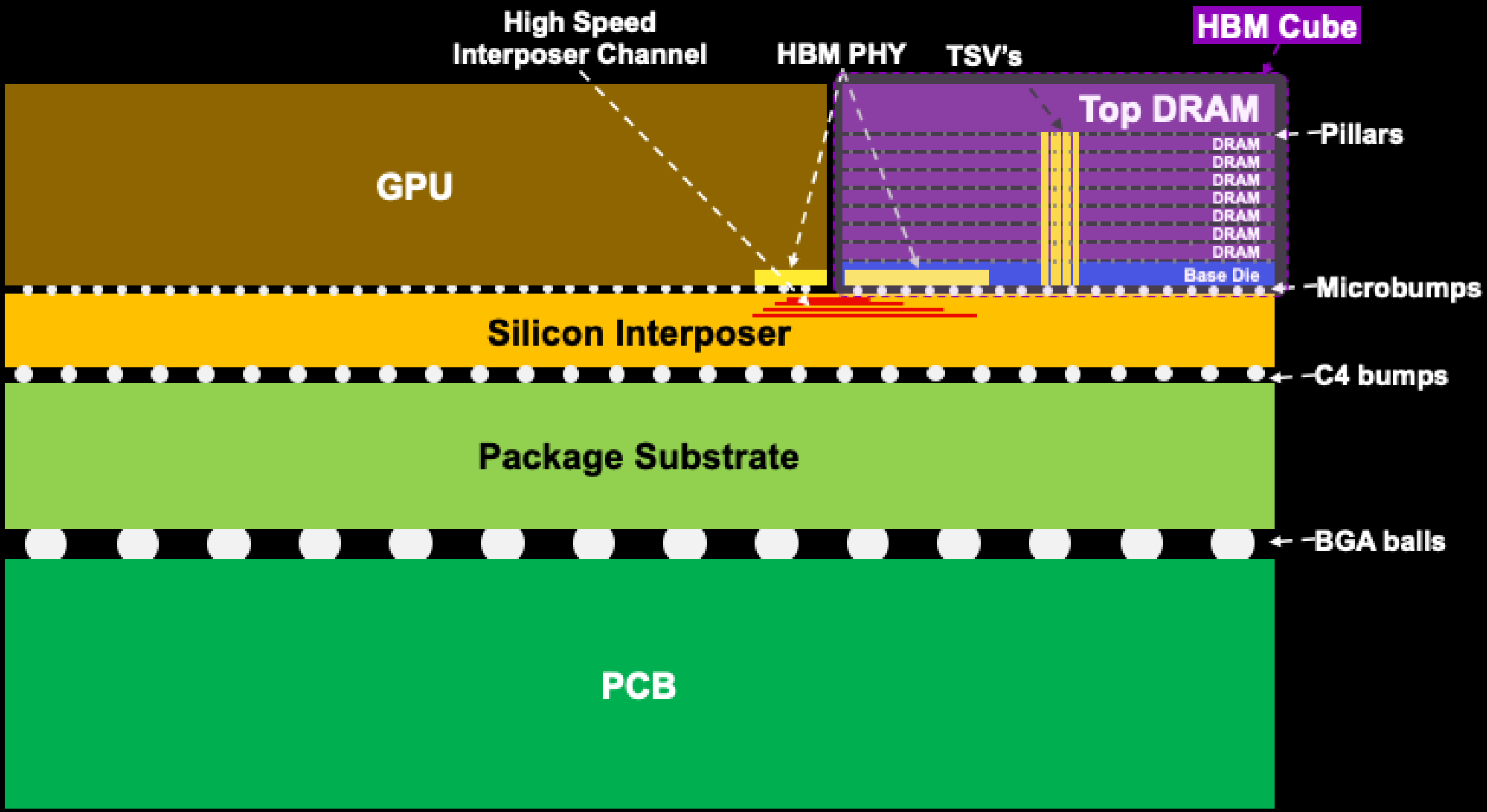
Product	HBM1	HBM2	HBM2E	HBM3	HBM3E	HBM4
Timeline	2014	2018	2020	2022	2024	2026
# Channels	8	8	8	16	16	32
# Pseudo-Channels (PC)	-	16	16	32	32	64
PC Width (bits)	128	64	64	32	32	32
Burst Length	2	4	4	8	8	8
Prefetch (bits)	256	256	256	256	256	256
Data I/O's	1024	1024	1024	1024	1024	2048
Nominal Data Rate (Gbps)	1	2.4	3.6	6.4	8	11
Nominal Bandwidth (GB/s)	128	307	460	819	1024	2800
DRAM Density (Gb)	2	8	16	16	24	24
DRAM Stack Height	4	4/8	4/8	8/12	8+	8+
Cube Capacity (GB)	1GB	4/8GB	8/16GB	16/24GB	24GB+	24GB+



Each generation increases:
- Data Rate → Higher bandwidth
- Pseudo-Channels → Granular & efficient
- Density & Stack Height → Larger capacity

HBM – Form Factor

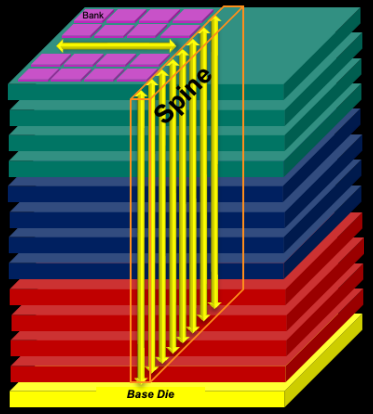
HBM integrated with GPU/Accelerator in a Heterogenous Integrated System – System-In-Package (SiP)

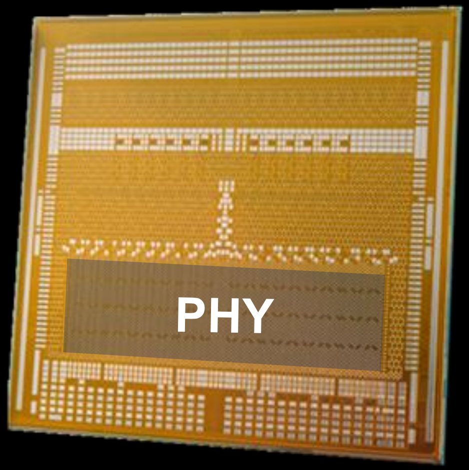


HBM Architecture

High parallelism through multiple independent channels

- Each DRAM die contains multiple independent channels
  - 2 Pseudo-channels per Channel
    - Shared Command/Address bus, independent data bus
  - HBM3E has 128 banks/die; HBM4 increases to 256 banks/die
- Base die interfaces command and data between host & DRAM
  - Microbump PHY connects the host to the base die
  - 3D TSV PHY interconnects the DRAM stack and the base die
- Highly dense & short point-to-point connections on interposer
  - 1K IO on HBM3E, 2K IO on HBM4





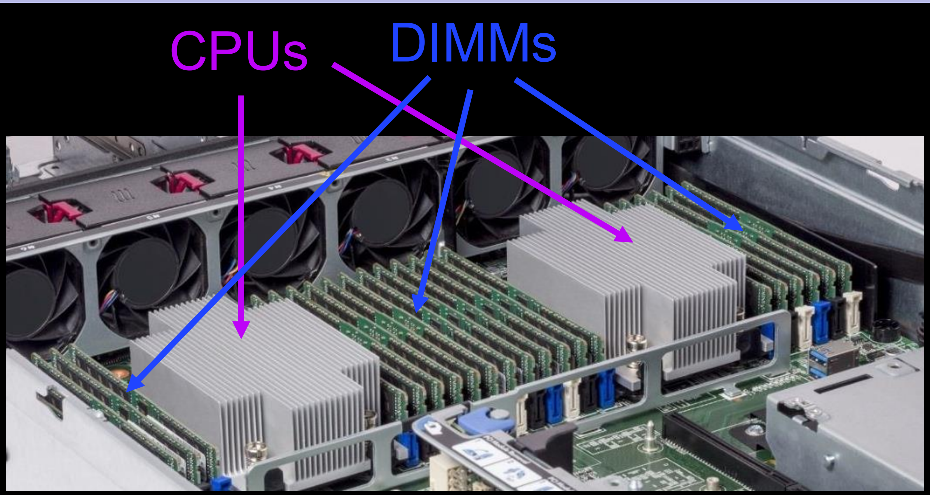
Source: Micron (HBM3E)

System Level Memory Bandwidth

HBM enables an order of magnitude higher system bandwidth versus DDR DIMMs

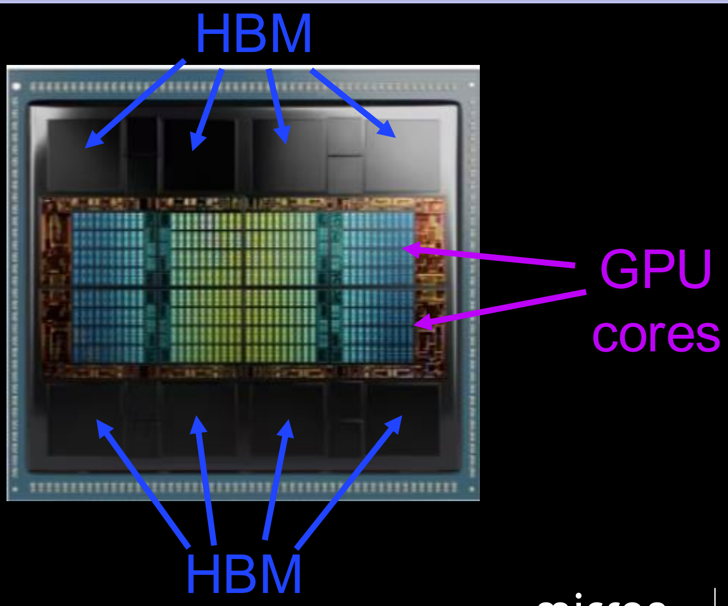
Servers have multiple DRAM channels per CPU to maximize memory bandwidth

- Example: 8 channels with DDR5 memory
  - 4800 MT/s, 2x32 bit sub-channels
    - ➢**307 GB/s theoretical bandwidth**
  - 2 ranks / channel, 64 GB / DIMM
    - ➢**1 TB capacity**



GPUs maximize bandwidth by using multiple HBM stacks

- Example: 8 HBM3 stacks at 5.2 Gbps
  - 16 channels per stack, 2 x32 bit pseudochannels
    - ➢ **5.3 TB/s theoretical bandwidth**
  - 24GB / stack
    - ➢ **192 GB capacity**



1. CPU: https://www.crucial.com/articles/about-memory/server-memory-and-hardware-compatibility
2. GPU: https://www.amd.com/en/products/accelerators/instinct/mi300/mi300x.html

System Level Memory Bandwidth

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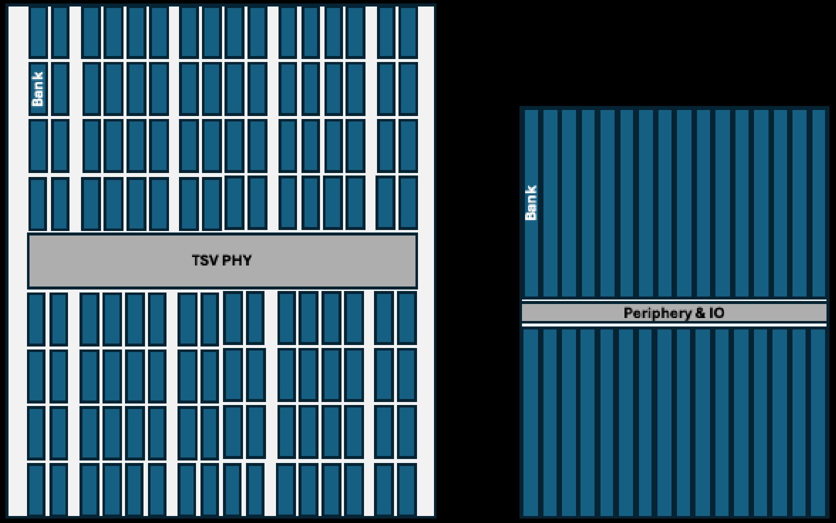
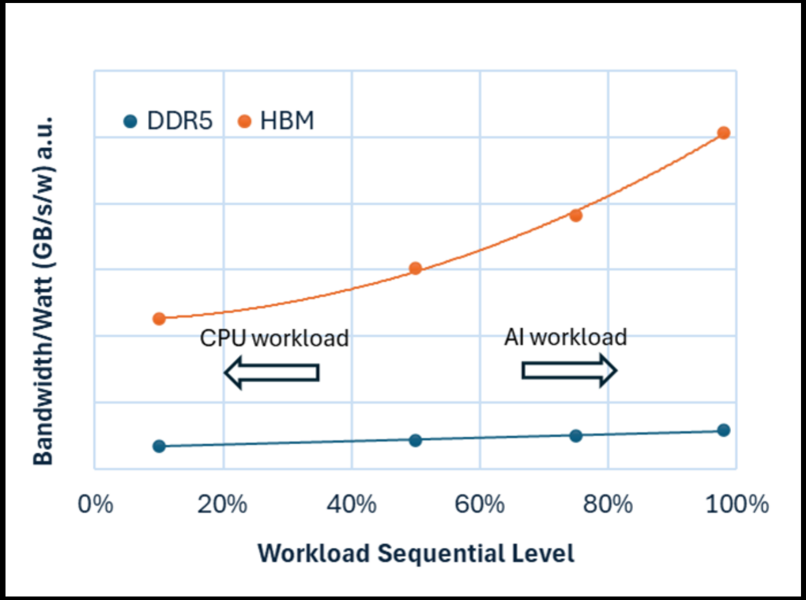


HBM3E
128 banks

DDR5
32 banks



Combined overhead from design architecture, advanced packaging, and manufacturing complexity results in ~**3X more silicon** being consumed for delivering HBM3E capacity relative to DDR5

	HBM3E	DDR5
Core BW @ 8Gbps	256GB/s = 256 IO*8Gbps	8GB/s = 8 IO*8Gbps
Banks	128 banks	32 banks

Essential memory parameters to support AI innovation

Performance

- Speed and bandwidth
- Reliability and fault tolerance

Capacity

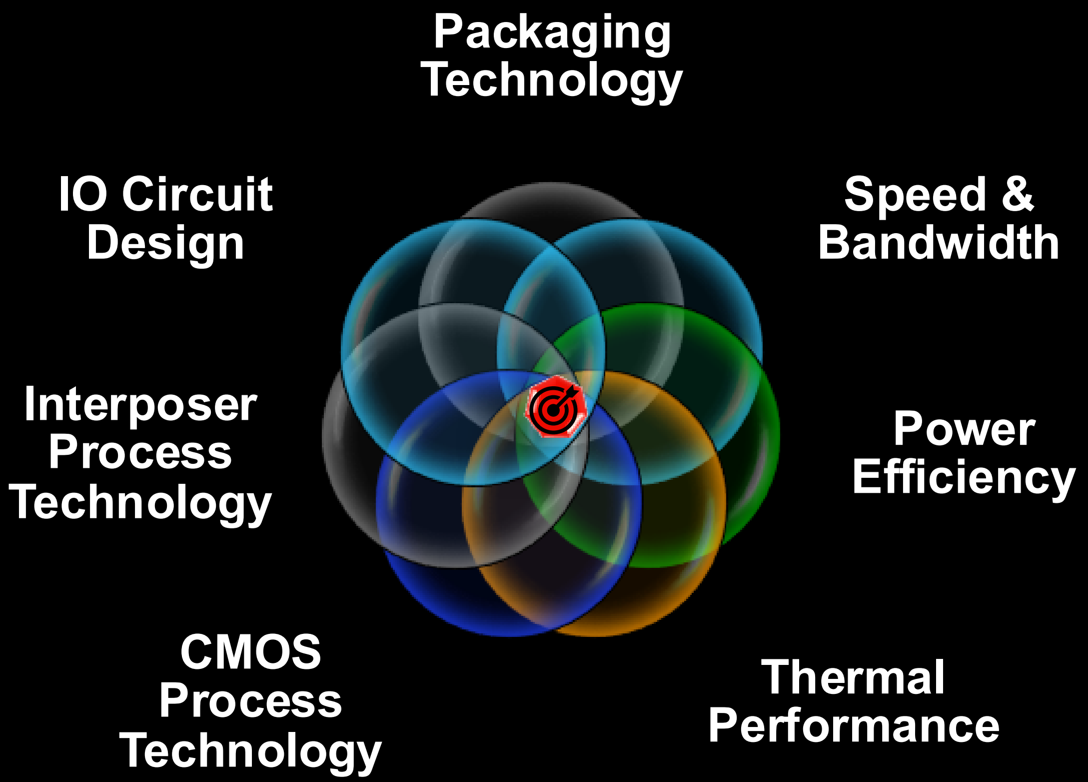
- Parameters, file and cache
- Scalability

Low power

- Energy efficiency
- AI-enabled edge devices

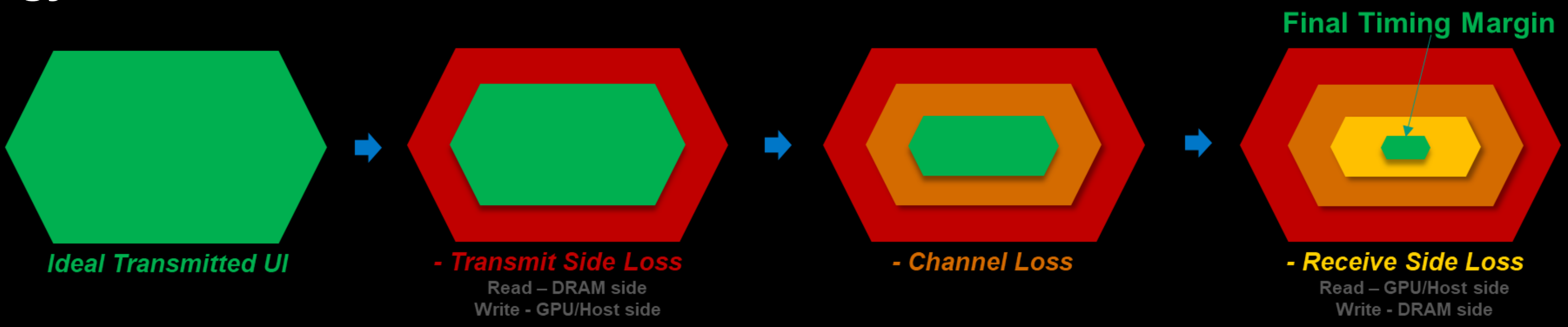
High Speed Links & SiP Integration

High-speed bandwidth scaling and larger interposers continue driving SiP advancements



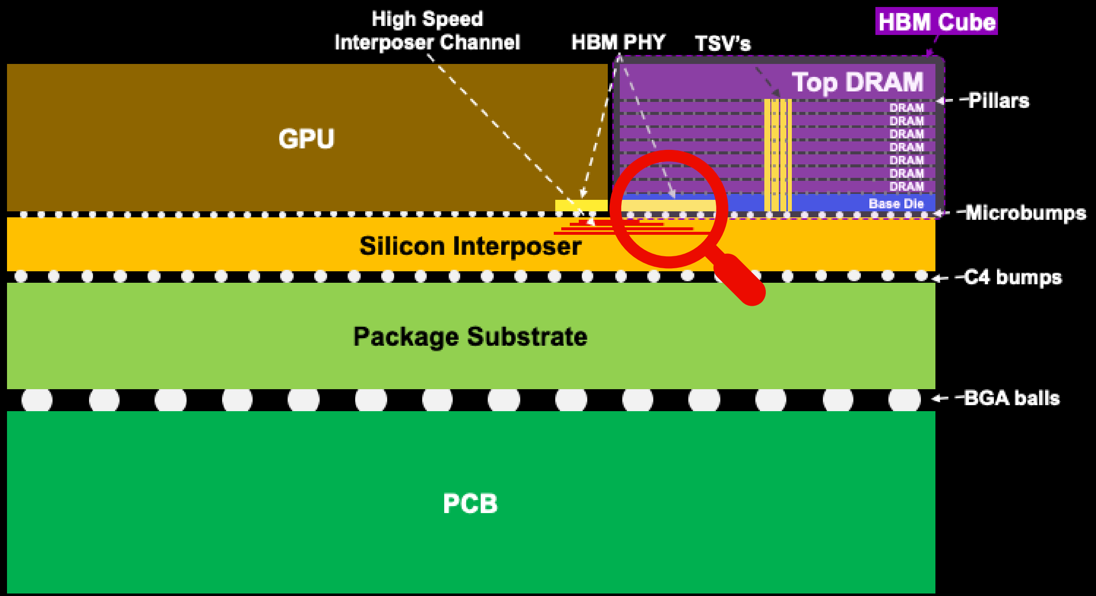
Higher speed IO links
- High-speed IO design innovation
- Memory-optimized SERDES D2D PHY
- CPO – Co-Packaged Optics

Larger interposer footprint
- Larger SiP's ~ CoWoS-L, CoWoS-R, etc
- Glass substrates



CPI & RAS – Key Challenges

CPI - Chip-Package-Interactions



Thermo-Mechanical pressure resulting from Coefficient of Thermal Expansion (CTE) mismatches between various materials used in complex heterogeneously integrated HBM device creates unique challenges

RAS - Reliability-Availability-Serviceability

Component	Category	Interruption Count	% of Interruptions
Faulty GPU	GPU	148	30.1%
GPU HBM3 Memory	GPU	72	17.2%
Software Bug	Dependency	54	12.9%
Network Switch/Cable	Network	35	8.4%
Host Maintenance	Unplanned Maintenance	32	7.6%
GPU SRAM Memory	GPU	19	4.5%
GPU System Processor	GPU	17	4.1%
NIC	Host	7	1.7%
NCCL Watchdog Timeouts	Unknown	7	1.7%
Silent Data Corruption	GPU	6	1.4%
GPU Thermal Interface + Sensor	GPU	6	1.4%
SSD	Host	3	0.7%
Power Supply	Host	3	0.7%
Server Chassis	Host	2	0.5%
IO Expansion Board	Host	2	0.5%
Dependency	Dependency	2	0.5%
CPU	Host	2	0.5%
System Memory	Host	2	0.5%

Table 5 **Root-cause categorization of unexpected interruptions during a 54-day period of Llama 3 405B pre-training.** About 78% of unexpected interruptions were attributed to confirmed or suspected hardware issues.

Two levels of ECC coverage starting on HBM3:

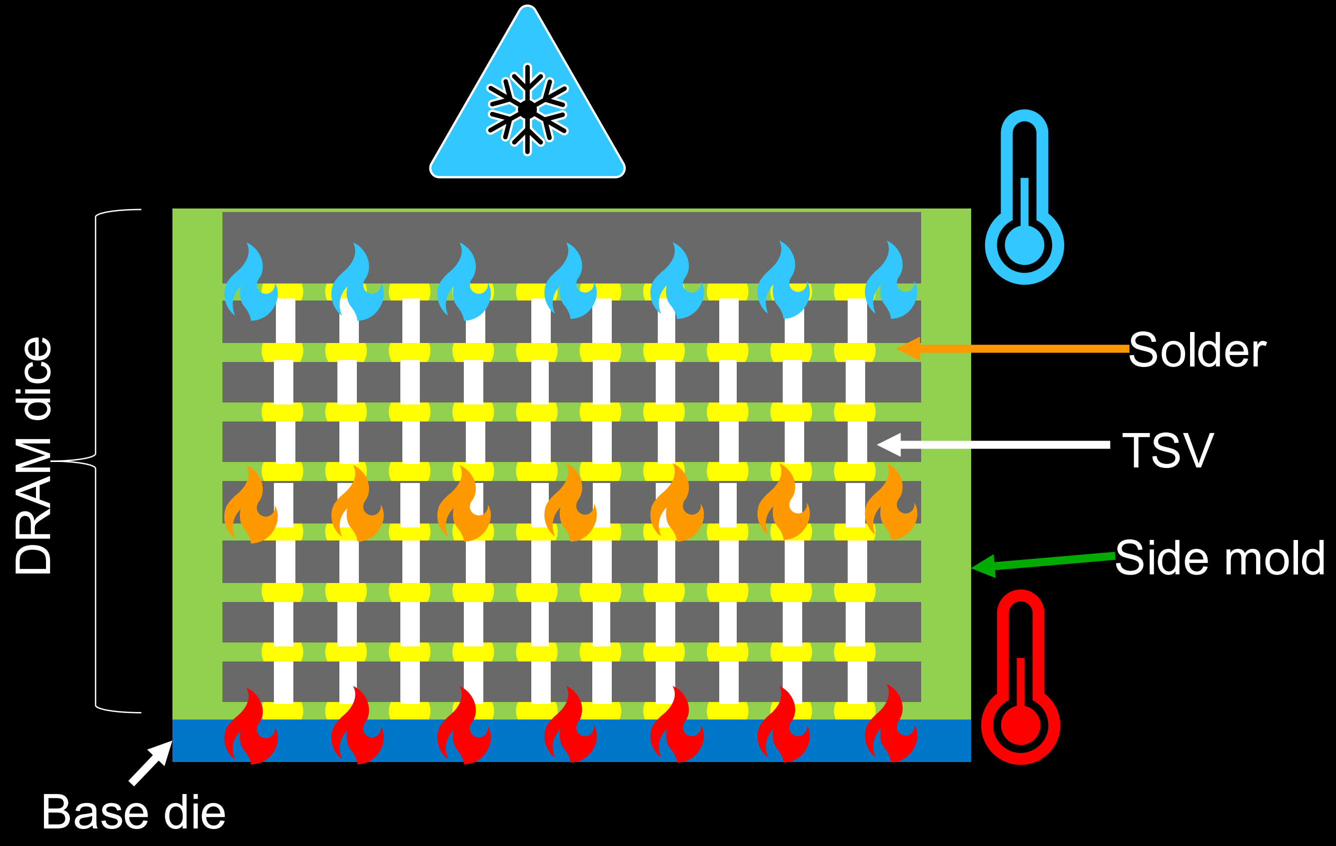
1. **System level coverage** – 16 meta bits per 256 bit access – system can deploy CRC or ECC based coverage
2. **On-die ECC coverage** – symbol based ECC (Reed-Solomon) implemented on-die

The Llama 3 Herd of Models, AI @ Meta, https://ai.meta.com/research/publications/the-llama-3-herd-of-models/

Thermal Challenges

Thermal innovations essential to enable continued bandwidth and capacity scaling

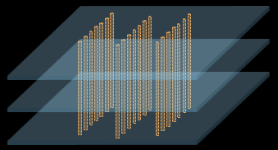
High Speed D2D Interconnect	🔥
Advanced functionality in base die	🔥
Increasing DRAM die activity	🔥
Increasing stack height	🔥
Liquid cooling	❄
Hybrid Bonding	❄



Advanced packaging industry trends

Diverse interconnects, stacking, and bonding technologies enable high-performance, low-power memory and logic integration

Drivers

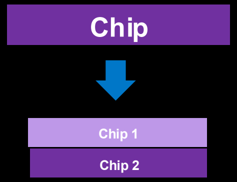


Area IO density scaling

ubumps, pads, TSVs

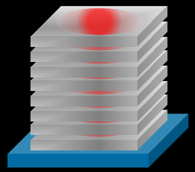
Lateral IO scaling

Lines and Spaces



Partitioning

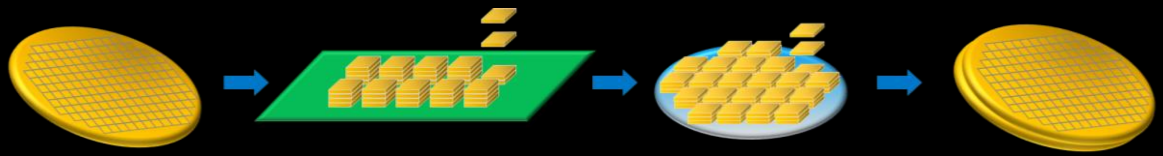
Array, CMOS



Thermals and PDN

Hot spots
Logic / Memory

Stacking technologies



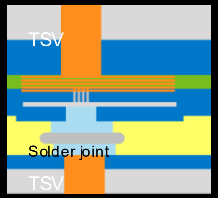
Wafer

CoS

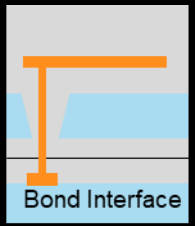
CoW/C2C/C2W

W2W

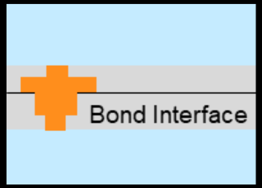
Interconnects



µbump



Fusion bonding



Hybrid bonding

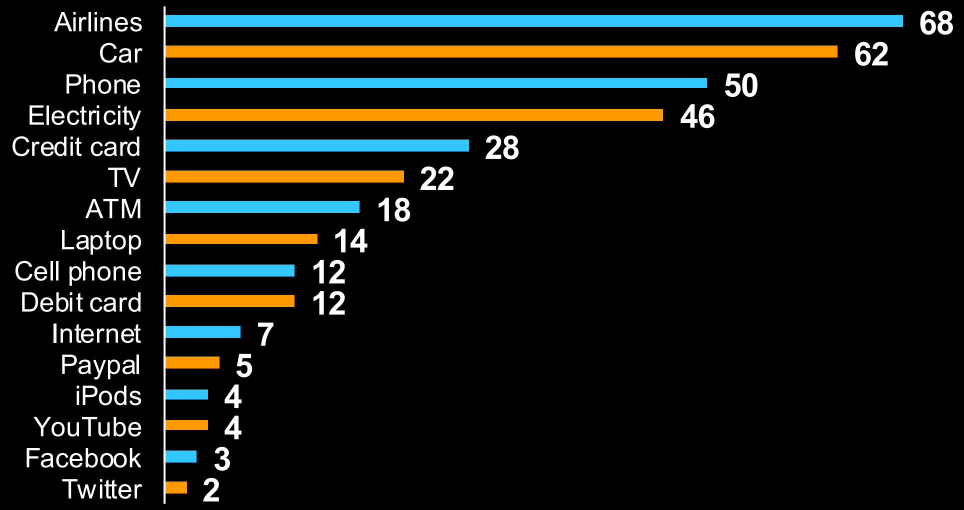
CoS: Chip on Substrate | **CoW**: Chip on Wafer | **C2W**: Chip to Wafer | **W2W**: Wafer to Wafer

Collaboration enables a stronger future for all

HBM turbocharging AI era

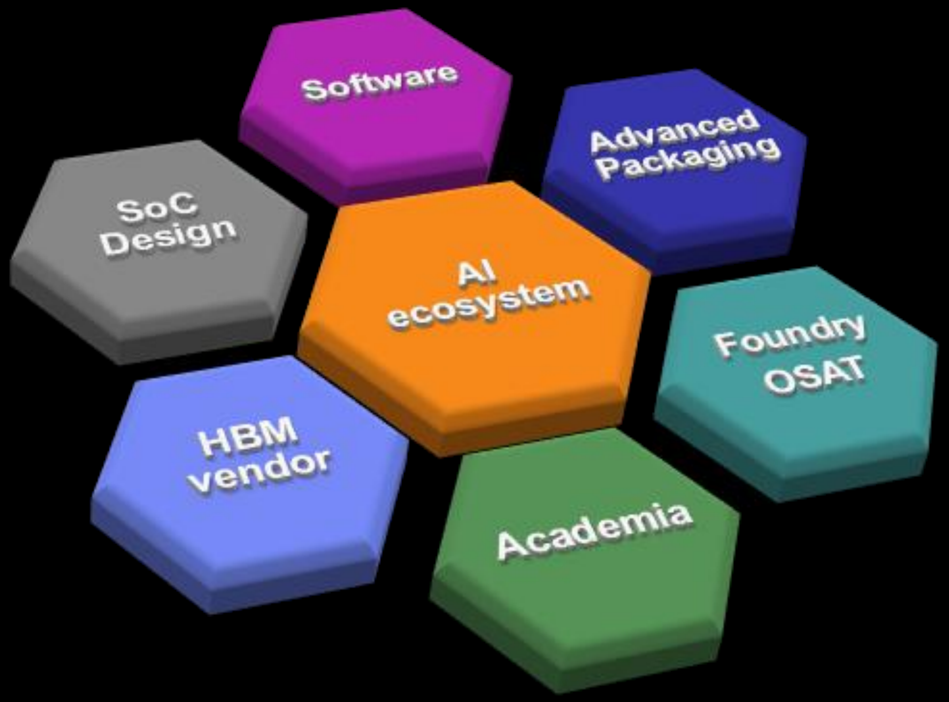
Unprecedented era

Number of *years* to get to **50 million** users[1]



Number of *months* to get to **100 million** users

ChatGPT ⟶ **2 months**



Close collaboration

1. https://www.futrcolab.com/blogs/pace-of-change-to-hit-50-million-users

Intelligence Accelerated